

STA301
QUIZ #3,4
FOR FINAL TERM
SPRING 2025
BY
BUTTERCUP
FOR LMS HANDLING
CONTACT: 0324-9427076

Quiz

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Hypothesis Testing Conclusio

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STA301 - Statistics and Probability (Quiz 4)

Quiz Start Time: 10:0

Question # 10 of 10 (Start time: 10:15:10 PM, 08 July 2025)

When the population is normally distributed, population standard deviation is unknown and the sample size is n = 15, the confidence interval for the population mean is based on

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Select the correct option

- the Binomial distribution.
- the Poisson Distribution.
- the z (standard normal)distribution.
- the t distribution.

Question # 9 of 10 (Start time: 10:14:50 PM, 08 July 2025)

The t-test for independent samples assumes which of the following?

Select the correct option

- Groups have equal means
- Groups have equal medians
- Scores are independent
- Scores are dependent

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STA301 - Statistics and Probability (Quiz 4)

Question # 8 of 10 (Start time: 10:13:11 PM, 08 July 2025)

For test of hypothesis Ho:μ1 = μ2 and H1:μ1 < μ2, the critical region at 0.05 level of significance and n > 30

Select the correct option

- Z<-1.6445
- Z>1.6445
- Z less and equal 1.96
- Z> 1.96

The correct option is B: Z>1.645.

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Question # 7 of 10 (Start time: 10:12:45 PM, 08 July 2025)

If a significance level of 1% is used rather than 5%, the null hypothesis is:

Select the correct option

- More likely to be rejected
- None of the these
- Just as likely to be rejected
- Less likely to be rejected

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The correct option is D:

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Question # 6 of 10 (Start time: 10:12:24 PM, 08 July 2025)

Critical region is also known as:

Select the correct option

- (●) Rejection region
- ( ) Neither a and b
- ( ) Acceptence Region
- ( ) None of these

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Question # 5 of 10 (Start time: 10:12:01 PM, 08 July 2025)

The term 1- β is called

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Select the correct option

- ○ Critical region
- ◉ Power of the test
- ○ Level of the test
- ○ Size of the test

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Tue 8 Jul 10:11 PM

STA301 - Statistics and Probability (Quiz 4)

Quiz Start Time: 10:08 PM, 08 July 2025

Question # 4 of 10 (Start time: 10:10:53 PM, 08 July 2025)

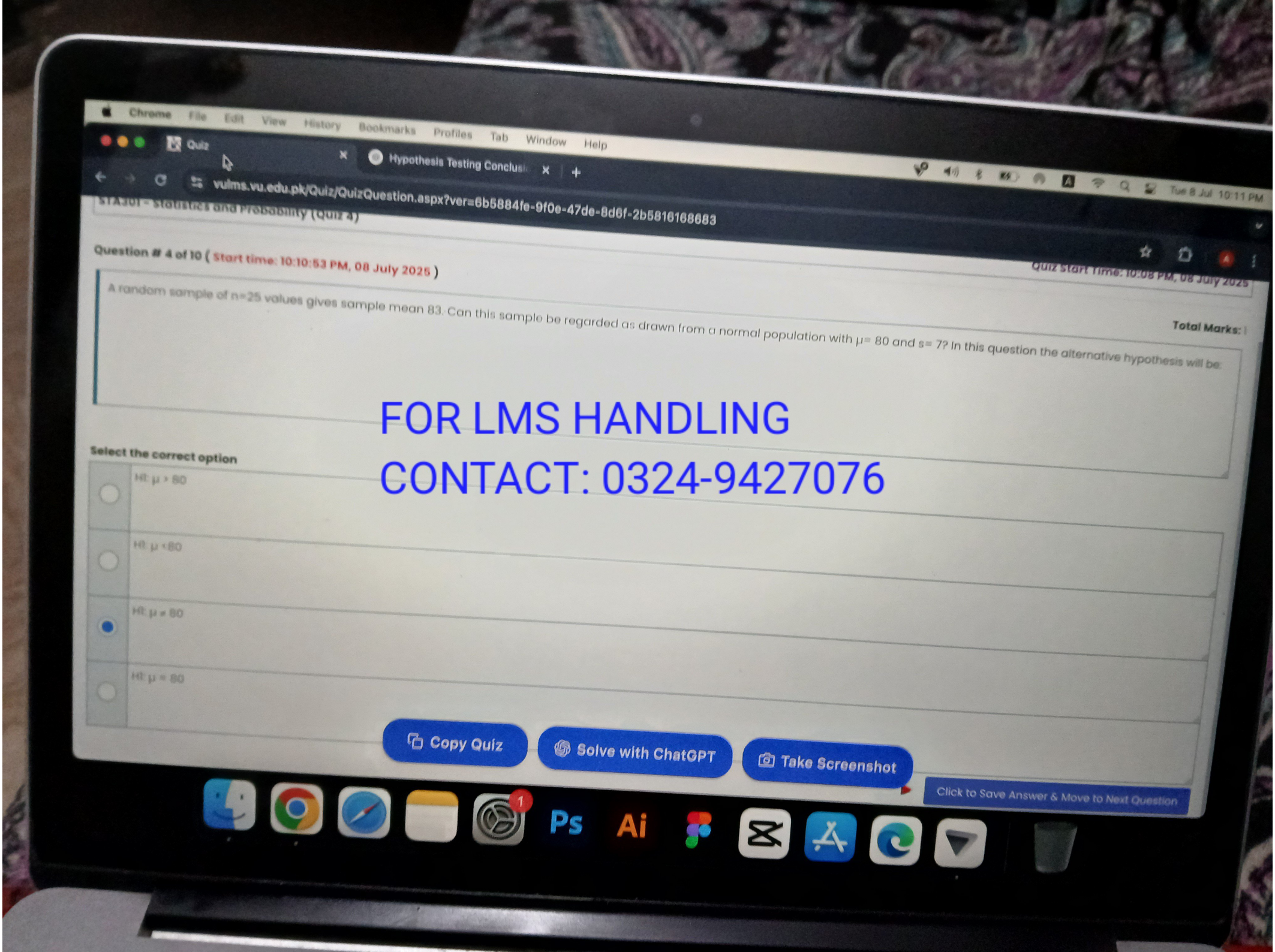
Total Marks: 1

A random sample of n=25 values gives sample mean 83. Can this sample be regarded as drawn from a normal population with μ= 80 and s= 7? In this question the alternative hypothesis will be:

Select the correct option

- H1: μ > 80
- H1: μ <80
- H1: μ ≠ 80
- H1: μ = 80

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STA301 - Statistics and Probability (Quiz 4)
Question # 4 of 10 (Start time: 10:10:53 PM, 08 July 2025)
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Select the correct option
H1: μ > 80
H1: μ <80
H1: μ ≠ 80
H1: μ = 80
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STA301 - Statistics and Probability (Quiz 4)

Question # 3 of 10 (Start time: 10:10:23 PM, 08 July 2025)

If type I and type II errors are fixed, then power of test increase with

Select the correct option

- ○ None of above
- ○ Changing the sample size
- ◉ Increasing the sample size
- ○ Decrerasing the sample size

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Time Left 70 sec(s)

Quiz Start Time: 10:08 PM, 08 July 2025

Question # 2 of 10 (Start time: 10:09:51 PM, 08 July 2025)

Total Marks: 1

If the NULL hypothesis says that the average height of Pakistani soldiers exceeds the average height of American soldiers by not more than 3 inches. What will be its ALTERNATIVE hypothesis?

Select the correct option

Reload Math Equations

- none of these
- P = 3
- P > 3
- P < 3

STA301 - Statistics and Probability (Quiz 4)

Question # 1 of 10 (Start time: 10:08:54 PM, 08 July 2025)

If we are testing H0: P1=P2 against H1:P1 > P2(or P1 -P2 > 0 and tabulated value of z = 1.96 and Z calculated value = 1.63 then what will be the conclusion:

Select the correct option

- None of the above
- Accept Ho
- Impossible to decide
- Reject Ho Reject H0

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The correct option is B: Accept Ho.

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STA301 - Statistics and Probability (Quiz 3)

Question # 10 of 10 (Start time: 07:20:36 PM, 24 June 2025)

Parameters of hyper geometric distribution are:

Select the correct option

○	5
●	3
○	2
○	1

Question # 9 of 10 (Start time: 07:20:06 PM, 24 June 2025) **Total Marks:** 1

The number of telephone calls that pass through a switchboard has a Poisson distribution with mean equal to two per minute. The probability that no telephone call passes through the switch board in a minutes is:

Select the correct option

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- (●) 0.1353
- ( ) 0.2707
- ( ) 0.0183
- ( ) 0.0517

Click to Save Answer & Move to Next Question

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Question # 8 of 10 (Start time: 07:19:23 PM, 24 June 202

E(4X + 5) =_____

Select the correct option

○	12 E (X)
○	16 E (X)
●	4 E (X) + 5
○	16 E (X) + 5

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Question # 7 of 10 (Start time: 07:18:52 PM, 24 June 2025)

If the second moment ratio is less than 3 the distribution will be:

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Select the correct option

- ○ None of these
- ○ Leptokurtic
- ◉ Platykurtic
- ○ Mesokurtic

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Question # 6 of 10 (Start time: 07:18:13 PM, 24 June 2025)

If c is a constant, then E(c) = _____

Select the correct option

○	0
●	c
○	-c
○	1

STA301 - Statistics and Probability (Quiz 3) **Quiz Start Time: 07:14 PM, 24 June 2025**

Question # 5 of 10 (Start time: 07:17:42 PM, 24 June 2025) **Total Marks:** 1

In a normal distribution how much area lies between $\mu \pm 2\sigma$

Select the correct option

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- (●) 95.45%
- ( ) 85%
- ( ) 94%
- ( ) 75%

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Question # 4 of 10 (Start time: 07:16:44 PM, 24 June 2025)

In sampling distribution of the sample proportion p is equal to:

Select the correct option

- ○ n/x
- ○ n
- ◉ x/n
- ○ x

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Question # 3 of 10 (Start time: 07:16:06 PM, 24 June 2025)

A random variable that can assume every possible value in an interval [a, b]:

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Select the correct option

○	Discrete variable
○	Qualitative variable
◉	Continuous variable
○	Categorical variable

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Question # 2 of 10 (Start time: 07:14:56 PM, 24 June 2025)

If b = 0, then E(ay+b) =

Select the correct option

○	y E(a+b)
◉	a E(y)
○	y E(a)
○	0

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Question # 1 of 10 (Start time: 07:14:01 PM, 24 June 2025)

Match the binomial probability P(x> 0.5) with the correct statement.

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Select the correct option

○	P(there are at most 0.50 successes)
○	P(there are at least 0.50 successes)
○	P(there are fewer than 0.50 successes)
◉	P(there are more than 0.50 successes)